

From Momentum to Meaning: Legislative Innovation and Impact in EU Policy Punctuations

Constantin Kaplaner*

May, 2025

This paper examines whether bursts of legislative activity — policy punctuations — within the European Union indeed translate into significant policy change, as suggested by punctuated equilibrium theory (PET). Drawing on a dataset of roughly 55,000 legislative documents, I develop a dual measure of legislative significance capturing both innovation (via advanced NLP-based content analysis) and impact (via citation network centrality). The results reveal that punctuated periods do not yield markedly higher levels of innovation and only modestly increase legislative impact, challenging the common assumption that heightened attention automatically fosters more groundbreaking or lasting policies. These findings refine PET by showing that not all legislative surges produce transformative outcomes; rather, many bursts result in incremental or routine measures. Consequently, the study underlines the importance of assessing what is legislated, rather than merely how much, and demonstrates the value of quantitative tools for distinguishing fleeting policy flurries from genuinely enduring reforms.

*LMU, Munich & Zeppelin University, Friedrichshafen | constantinkaplaner@gsi.uni-muenchen.de

1 Introduction

Policy change is often conceptualized not as a steady, incremental process, but rather through significant shifts known as policy punctuations. This concept is central to the punctuated equilibrium theory (PET), originally developed in evolutionary biology by Gould and Eldredge (1977), and later adapted to political science by Baumgartner and Jones (1993). PET suggests that political systems are usually characterized by long periods of stability (equilibrium), interspersed with short bursts of substantial change (punctuations). These punctuations disrupt the status quo, potentially leading to major changes that reshape policy landscapes.

The theory posits that political and institutional structures typically resist change due to various stabilizing mechanisms, such as entrenched interests, institutional friction (Baumgartner et al. 2009), and bureaucratic inertia (Robinson et al. 2007). However, when these stabilizing forces are overcome — often by external shocks, shifts in public opinion, or political realignments — policy-making can undergo rapid and dramatic transformations.

Often assessed in terms of change rates between periods, a core assumption of PET is that shifts in these rates signify *significant* departures. This assumption is understandable, as PET literature has historically focused on changes in budget distributions (Kuhlmann and van der Heijden 2018). In this context, significance is directly tied to the subject matter, as it reflects alterations in resource allocation (Adolph, Breunig, and Koski 2020). Similarly, qualitative studies of policy punctuations often examine specific instances precisely *because* of their perceived significance. However, when examining punctuations in the distribution of legislative outputs, this assumption of significance is not always valid. Legislative outputs can vary greatly in scope and relevance, meaning that a punctuation in legislative output might be purely symbolic or procedural (Givel 2006; John and Bevan 2012). See also Mortensen (2009), who emphasizes that surges in political attention do not always yield proportionate shifts in policy content or significance, even when outputs spike. Some scholars have attempted to address these issues by introducing qualifying measures to weight legislative outputs in advance, either by the structural extent of the legislation (Beaulieu-Guay, Costa, and Montpetit 2023) or by focusing directly on policy content (Fernández-i-Marín et al. 2019; Hurka, Adam, and Knill 2017). While the latter often involves dissecting policy content manually, making large-scale studies less feasible, the former approach often fails to capture the true substance of the policy (Hurka, Knill, and Steinebach 2023).

Consequently, the question of whether a punctuation in legislative outputs captured through

quantitative patterns alone can be seen as significant change remains mostly unanswered. Exemplary for this is the discourse between Hall (1993) and Baumgartner (2013). In his seminal work Hall (1993) distinguishes between three types of policy change: first-order (changes in settings or instruments), second-order (changes in techniques or strategies), and third-order (changes in goals or paradigms). These attempts to give nuance to the perspective of change have been mostly forgone by the PET literature — partly because of the difficulty of applying them on a large scale partly due to the assumption that overall change patterns capture at least aspects of these more granular views (Baumgartner 2013). This potential disconnect between macro level pattern and micro level significance might have led to a plethora of proposed typologies of policy change based on various criteria to (re)introduce some nuance in the study of policy change (Cashore and Howlett 2007; Howlett and Cashore 2009; Bauer and Knill 2014).

This paper addresses this question by investigating the relationship between policy punctuations and legislative significance, with a specific focus on the European Union (EU). The EU is a compelling case for studying policy punctuations and legislative significance because, much like national political systems, it operates as a complex political entity with decision-making authority across a wide range of domains. As Lundgren, Squatrito, and Tallberg (2018) point out, policy-making in this setting — like domestic contexts — is subject to constant societal pressures, such as economic crises, technological shifts, and environmental challenges. The EU, with its comprehensive policy-making scope, faces similar demands for policy responses, just as national governments do. Moreover, the EU’s institutional structure, with its multi-level governance, is prone to similar types of cognitive and institutional barriers that characterize national political systems (Scharpf 1988; Tsebelis and Yatağan 2002). These capacity limitations create the conditions for policy punctuations, aligning closely with PET (see also Princen 2013).

Legislative significance is defined here as the functional importance of a policy, operationalized through two dimensions: innovation and impact. Innovation refers to the degree to which a legislative output deviates from existing frameworks, while impact assesses the long-term influence of a policy within a broader legislative network. This dual approach allows for a more rigorous evaluation of whether periods of policy punctuations result in substantial and enduring policy changes, or merely temporary shifts in output volume. Methodologically, this research introduces a novel approach for measuring legislative significance. Drawing on roughly 55,000 legislative documents, the study utilizes state-of-the-art large language model (LLM) techniques to assess the innovation dimension and network-based

analyses to capture the impact of legislative outputs.

The findings of this study challenge the assumption that policy punctuations as captured through activity shifts inherently reflect more innovative legislation. While policy punctuations are associated with slightly increased legislative impact, suggesting that laws enacted during these periods tend to have long-term significance, the results indicate that these periods do not consistently produce higher levels of innovation. This decoupling between legislative output and innovation suggests that policy makers may prioritize consolidation and incremental adjustments during periods of heightened legislative activity, rather than pursuing fundamentally new approaches.

In doing so, this paper contributes to the broader theoretical understanding of punctuated equilibrium in policy-making, offering new insights into the conditions under which bursts of legislative activity translate into substantive policy change. By combining micro level and macro level perspectives on legislative outputs, this research provides a more comprehensive framework for evaluating the significance of policy punctuations, both in terms of immediate innovation and long-term impact. These findings have implications not only for the study of policy change but also for policy makers seeking to understand the drivers of meaningful legislative reform.

The paper is structured as follows. Section 2 outlines the theoretical considerations on how political systems operate during significant shifts in attention and explains why these shifts might influence the significance of the legislation produced. Section 3 explains how legislative significance can be conceptualized and measured in terms of innovation and impact, describes the data used for this analysis, and details the estimation approaches employed to assess the association between significance and punctuations. Section 4 presents the empirical findings, and Section 5 discusses their implications for categorizing policy punctuations. Finally, Section 6 concludes the paper and examines the broader implications for policy change.

2 Theory

2.1 Understanding policy punctuations

One persistent challenge in the study of policy change is defining what constitutes significant change and understanding what different measures reveal about it. This issue has led to critiques of PET, notably by Howlett and Cashore (2009), who argue that assessing

change solely through shifts in legislative activity oversimplifies the diverse patterns of policy evolution.

PET posits that political agendas are stable for extended periods but occasionally undergo large-scale changes — punctuations — when issues shift from routine decision-making to the forefront of political attention (Baumgartner and Jones 1993; Baumgartner et al. 2009). While PET emphasizes these bursts of activity, relying solely on quantitative measures of legislative output can obscure the mode of change — the fundamental nature of the policy shift.

The distinction between the tempo and the mode of change is crucial, as discussed by Howlett and Cashore (2009). Tempo refers to the speed or magnitude of change, while mode pertains to the nature or type of change — whether it is an incremental adjustment within existing frameworks or a fundamental shift in policy paradigms (Durant and Diehl 1989). This notion links to Hall (1993) who in similar vein distinguishes between first-order changes (adjustments in policy settings), second-order changes (modifications of policy instruments), and third-order changes (transformations of overarching policy goals and paradigms).

Capturing the mode of policy change is difficult for several reasons. First, aggregate data often do not reveal whether changes are incremental tweaks or fundamental transformations (John and Bevan 2012). Quantitative measures can indicate that large activity shifts occur but not the underlying nature of these. Second, different policy areas have their own definitions and thresholds for what constitutes significant or paradigmatic change. A major shift in one policy domain might be considered minor in another, complicating cross-comparison and the development of a unified framework. Moreover, changes in tempo might not necessarily correspond to changes in mode; rapid changes can still be incremental, and fundamental shifts can occur gradually, making classification challenging.

One potential solution to this problem is adopting the concept of legislative significance. Mayhew (2005) suggests that legislation is significant if it is “both innovative and consequential — or if, at the time of passage, it is thought likely to be consequential.” This introduces two critical dimensions: innovation — the extent to which legislation departs from established norms, introducing new frameworks or challenging the status quo — and impact — the long-term influence a law exerts within the broader system, which may occur even without substantial innovation.

Legislative significance differs from the concept of mode in important ways. While mode focuses on categorizing the *nature* of change, legislative significance assesses the *importance*

of a policy change based on its observable outcomes — innovation and impact — without needing to pre-define the type of change. By evaluating legislation based on these tangible outcomes, researchers can apply this framework across different policy subsystems, bypassing the issue of varying definitions of significant change.

2.2 The connection between attention shifts, innovation, and, impact

Before exploring how to measure innovation and impact, it is crucial to first assess whether these concepts are truly independent from shifts in legislative activity. If shifts in legislative activity consistently resulted in increased innovation and impact, separate analysis of these dimensions would be redundant. However, both empirical evidence (see [John and Bevan 2012](#)) and theoretical considerations suggest that attention shifts do not always translate into innovation or impact in legislative outputs.

A key reason why shifts in legislative activity might not always result in innovative or impactful legislative outputs might lie in the complex way information is processed during the decision making stages. Padgett ([1980b](#))’s hierarchical garbage can model offers a useful framework for understanding how organizations handle information and make decisions. Building on earlier work by Cohen, March, and Olsen ([1972](#)), who developed the garbage can model to describe decision-making in organizations characterized by ambiguity and chaotic environments, the hierarchical garbage can model adds layers to this process, demonstrating how decision flows are filtered and condensed at different levels of authority within an organization. While Padgett ([1980b](#))’s hierarchical garbage can model offers a valuable perspective on how organizations process information and make decisions, it does not account for many of the broader political complexities involved in policy-making. The model focuses primarily on organizational structures and decision flows, but policy-making is also shaped by political negotiation, power dynamics, and institutional constraints that extend beyond internal decision processes ([Baumgartner and Jones 1993](#); [Kingdon 1984](#)). These factors complicate the connection between attention shifts and significant legislative change, as legislative outcomes are rarely determined by information processing alone. Instead, they emerge from a dynamic interplay of actors, interests, and institutional frameworks ([Sabatier 1999](#); [Hall 1993](#)), making the translation of punctuations into legislative outputs more nuanced than organizational models can fully capture. Nevertheless, it offers an analytical tool to at least capture certain areas that might influence the connection studied here.

Thus, following the model, decision-making is structured in tiers. At the lower levels, minor

agents monitor and condense incoming environmental information. Program chiefs then evaluate this information and suggest changes, which can range from minor adjustments to major departures from the status quo. Minor changes are typically rubber-stamped by higher authorities, Padgett (1980b) uses the terms secretaries and presidents, while more significant changes undergo additional scrutiny before being acted upon. While Padgett (1980b) was concerned with budgeting decisions, the same rational might be applicable to the process of policy formulation. Hence, employing this perspective of decision flows might help to explain why shifts in legislative productivity do not always lead to significant legislative changes across the dimensions of innovation and impact. Even when shifts signal important new issues, these issues are filtered through layers of decision-making where institutional routines and established processes might dominate decision making (Cyert and March 1963). Consequently, there are two major components to this model. First, the actors within it are bounded rational and, second, the hierarchical structure actors are embedded in.

Regarding the first, even when major issues emerge and are recognized by higher levels of decision-makers, the policy response may still be constrained by the bounded rationality of policy-makers, who are limited in their capacity to process complex information comprehensively (Simon 1957; Padgett 1980a). This might result in the continuity of institutional routines and reliance on known regulative approaches. In chaotic information environments, policy-makers cope with complexity and time limitations by structuring their perceptions of issues within “task environments” (Payne 1982). These environments consist of belief systems, relationships, and institutional rules that simplify the search for information by guiding decision-makers toward trusted sources. Rather than conducting an exhaustive search, policy-makers might employ heuristics, focusing on familiar, reliable sources of information (Gigerenzer and Selten 2002). As long as the outcomes are satisfactory, these heuristics are repeated, reinforcing risk-averse behaviors that favor incremental over radical change (Holyoke and Brown 2019). Especially in situation where attention drastically shifts, policy-makers might rely on these routines to craft legislation, making significant departures less likely.

Furthermore, policy-makers are limited by cognitive and informational constraints, which lead them to focus on a small number of issues at any given time. A factor that is also reflected in the vertically and horizontally segregated institutional structure they operate in. When focus shifts dramatically due to drastic shifts in activity, the processing of this new information often prioritizes immediate, visible problems without necessarily reevaluating the underlying policy framework. As Padgett (1980b) suggests, decisions flow upwards

more frequently when the informational environment is volatile or when the range of issues expands, but this upward flow does not guarantee that the broader agenda will be altered. Instead, the system may still rubber-stamp minor changes and reserve deep scrutiny only for cases where the departure from the status quo is so large that it forces a reassessment of priorities. Both these factors should primarily influence the potential for crafting innovative legislation in these periods.

Secondly, the hierarchical nature of decision-making introduces further delays or constraints on significant change. Major departures from the status quo undergo extensive review and often face resistance at higher levels, where policy-makers are more risk-averse and focused on maintaining institutional stability. As a result, while shifts in attention at lower levels may generate significant activity, they may ultimately translate into many incremental legislative changes rather than substantial reforms.

While some scholars, such as Baumgartner and Jones (1993), argue that punctuations occur when policy monopolies break down or issues escalate beyond their subsystem onto the macro-political agenda—creating opportunities for rapid and sweeping reforms—such instances might be rare. The persistence of entrenched interests, combined with the bureaucratic inertia within institutions (Robinson et al. 2007), means that policy monopolies often recover quickly or do not breakdown (Givel 2006), reasserting or remaining their influence and reinforcing the status quo. Even when a window for change opens, external pressures or crises may be required to sustain momentum for more radical shifts. Without such conditions, the system might revert to a pattern of incrementalism, where changes are minor, and the broader structure of policy remains intact (Lindblom 1959). Hence, it is likely that activity shifts observed do not align with innovative legislative changes.

Yet, it is important to note that this perspective primarily influences the innovation dimension of legislative significance. When it comes to impact, however, the dynamics can differ. Shifts in attention often result from problem pressures — situations where external crises or long-ignored issues suddenly demand urgent action (Fernández-i-Marín et al. 2019). In such cases, the priority may not be innovation but rather addressing immediate needs through legislation that consolidates or strengthens existing frameworks. This can lead to policies that, while not necessarily innovative, have significant long-term impact due to their role in stabilizing or reinforcing the legislative system.

For example, even if the solutions implemented during periods of attention shifts are incremental or rely on familiar approaches, they may still reshape the policy environment by influencing subsequent legal frameworks, establishing new regulatory norms, or integrating

into larger governance structures. This enduring relevance, measured through impact, shows that activity shifts, even when not characterized by high innovation, can produce legislation with considerable staying power. This legislation can ripple through the system by creating precedents or embedding practices that shape future policy-making. Thus, while innovation may be constrained by the hierarchical and cognitive limitations of decision-making, the impact of policies passed during these punctuated periods can still be profound, ensuring that these laws remain central to governance long after the punctuations have subsided ([Holyoke and Brown 2019](#)).

Thus, while punctuations reflect moments when attention shifts beyond a critical threshold, the way that information is processed within institutions — through layers of decision-making and bounded rationality — might limit the ability of these shifts to translate into comprehensive policy reforms. Technocratic policy formulation, especially in hierarchical structures, might tend to prioritize routine and incremental adjustments, reducing the likelihood that innovative legislative outputs will result from even substantial attention shifts. Therefore, while punctuations may lead to greater impact, they are less likely to foster the innovation necessary to break away from established norms.

3 Research design

3.1 Capturing legislative significance

As established before, I conceptualized legislative significance as a two dimensional construct consisting of the innovation and impact of a legislative output. When it comes to measuring these dimensions, most approaches tend to focus on one of these two dimensions. Clinton and Lapinski ([2006](#)) provide a comprehensive overview of prior efforts to map legislative significance in the United States, revealing two major insights. First, nearly all existing approaches rely primarily on subjective assessments rather than objective measures of significance (see also [Adler and Wilkerson 2012](#)). Second, as a result, these subjective measures are often difficult to compare over time and across different sources, leading to ambiguous criteria for determining legislative significance and defining its thresholds. The same challenge of subjective and fragmented assessments of legislative significance has been observed in EU policy-making when comparing subjective assessments with different objective indicators ([Warntjen 2012](#)). To address these limitations and in line with recent

research by Zubek, Dasgupta, and Doyle (2021), I will use objective measures that capture legislative significance across both dimensions.

3.2 Measuring innovation

To measure innovation, I define it as the extent to which a policy deviates in content from previous policies. I will focus specifically on the enacting terms of a legislation, rather than the entire text, which in the case of EU legislation may include recitals and annexes. My objective is to capture the core substance of the legislative output — the specific actions, regulations, or provisions that will be implemented. The enacting terms represent the actionable components of a legal text and are most likely to indicate innovation by introducing new regulatory measures, altering existing frameworks, or shifting policy direction. In contrast, recitals and annexes, while providing context and supplementary details, generally do not contain the essential legislative changes or mandates. Instead, they elaborate on the background, purpose, or technical specifications, which may not directly reflect the policy’s innovative aspects (Den Heijer, Abeelen, and Maslyka 2019). By concentrating on the enacting terms, I aim to assess the novelty of the policy’s practical and enforceable elements, ensuring a more precise and relevant measure of innovation.

The enacting terms are first divided into articles, which will serve as the primary units of analysis. Each article is further split into individual sentences using the *en_core_web_sm* from *spaCy*. These sentences are then embedded into a high dimensional space using S-BERT with the *all-mpnet-base-v2* model (Reimers and Gurevych 2019), a LLM that generates sentence embeddings. For each article, the sentence embeddings are averaged to produce an article-level embedding, capturing the unique content and nuances of that article. This approach allows for a detailed numeric representation of the policy’s content at the article level. Unlike traditional text-as-data approaches, such as the bag-of-words model that converts documents into vectors of word counts (see Grimmer and Stewart 2013), the use of S-BERT allows for a more nuanced, high-dimensional representation that better captures the semantic content of each article. Figure 1 visualizes the resulting embeddings for all EU legislation stemming from 1980 till 2015¹. For this, the high dimensional embeddings are reduced to three dimensions using UMAP (McInnes, Healy, and Melville 2018). Each point represents an article, with colors corresponding to the policy field as classified by the EuroVoc indicators (see Kaplaner, Knill, and Steinebach 2025) it originates from. Articles with similar content are positioned closer together, while those with differing content

¹For more information on the exact data used see Section 3.5.

are placed farther apart. As expected, articles from the same policy field tend to cluster together, reflecting their thematic similarities.

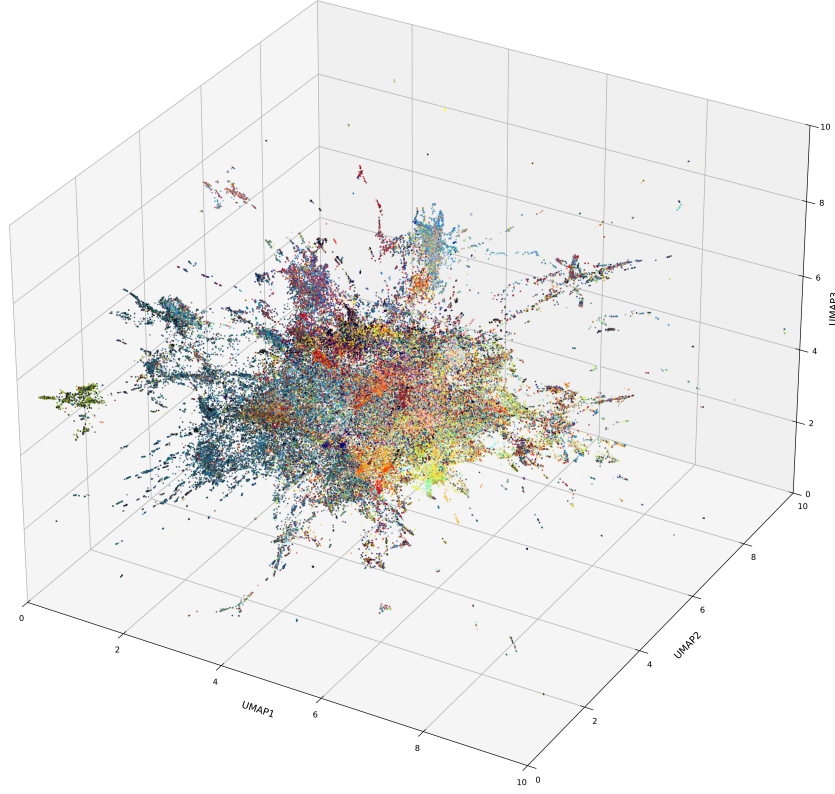


Figure 1: UMAP representation of article embeddings. $N = 356,012$. Each point represents one article. Each article is colored according to the corresponding policy field based on the EuroVoc classification scheme.

But how does this approach help assess the innovativeness of a policy? To address this, I leverage the fact that similar articles are clustered together while dissimilar ones are positioned apart. As previously mentioned, innovativeness can be understood as a departure, or in this case, the distance from the status quo. By using high-dimensional embeddings, it

becomes possible to evaluate how far an article deviates from prior legislation. Consequently, the innovativeness I of an article A can be defined as the average cosine distance between the article and previous articles S . This is formalized as:

$$Innovation_{j,i} = I_{j,i} = \frac{1}{n} \sum_{t-x}^{t-1} \sum_k \left(1 - \frac{A_{j,i} \cdot S_{i,t,k}}{\|A_{j,i}\| \|S_{i,t,k}\|} \right) \quad (1)$$

Where the innovativeness I of a policy j from sector i at time t is calculated as the mean cosine distance between the article and prior articles. I calculate the innovation score once with regards to all prior legislation (see formula above) and once only comparing legislation within the same field. The reference time frame is defined by x , with the default being $x = 5$ ². The assumption is that the further a policy deviates from previous policies, the more innovative it is.

3.3 Measuring Impact

To evaluate the impact of a policy, I will utilize the highly referential nature of laws, enabling their study through a network perspective (Bommarito, J, and Katz 2010; Fjelstul 2019). In this framework, each law is represented as a vertex, with edges between vertices indicating references to other laws. These edges can be either incoming (being cited by another law) or outgoing (citing another law).

While it may seem intuitive to assess the significance of a law based on its inward citations — how frequently it is referenced by other laws (Conti-Brown and Ohlrogge 2021) — there is also a strong argument for considering outward citations. Fowler et al. (2007) assesses legal importance of precedents at the U.S. Supreme Court. Additionally to inward important cases — cases that are referenced often — they also consider outwardly important cases. An outwardly important case, for instance, might reference numerous other cases to determine which rulings apply to a specific legal issue (Fowler et al. 2007), or it might overturn or alter previous rulings. Although this framework focuses on legal cases, the same logic could be transferable to legislation. Laws that cite many other laws may serve to clarify or modify existing provisions. Thus, both inward and outward citations are important for evaluating a law’s impact.

²I select $x = 5$ as it strikes a good balance, providing enough comparisons with previous periods while minimizing data loss. As a result, the number of articles for which innovation scores can be calculated is reduced from 356,012 to 344,768. See also Section 3.5.

Consequently, I will not distinguish between outward and inward relevance and focus on the total impact of an law. From a technical perspective, this approach implies capturing the legal landscape as an undirected network, where connections between laws are recorded without distinguishing the direction of the citation.

To capture the impact of a law, I will focus on eigenvector centrality as the primary measure of importance. Unlike simpler measures such as degree centrality, which only counts the number of direct connections a law has, eigenvector centrality provides a more comprehensive view by considering not only the number of connections but also the influence of the connected vertices themselves (Wasserman and Faust 1994). This measure captures the idea that connections to highly influential vertices contribute more to a law’s relevance than connections to less influential ones. Eigenvector centrality is formalized as:

$$Impact_{j,i} = x_{i,j} = \frac{1}{\lambda} \sum_j A_{ij} x_j \quad (2)$$

In this formula, $x_{i,j}$ represents the eigenvector centrality of vertex (law) j from a policy field i , λ is the eigenvalue corresponding to the eigenvector, A_{jk} is the element of the adjacency matrix A representing the connection between vertices j and k , and x_k is the eigenvector centrality of vertex k . The recursive nature of this formula ensures that a law’s centrality increases not just by having many connections, but by being connected to other highly influential vertices. I choose eigenvector centrality over simpler measures like degree centrality because it captures a law’s systemic influence more accurately. By accounting for the influence of connected vertices, eigenvector centrality better reflects the law’s importance within the network, making it a more robust indicator of legislative impact.

Figure 2 illustrates the differences between the two centrality measures for a simple network. Vertices located at the outskirts of the network tend to have a degree centrality of one, meaning they have only a single connection. However, their eigenvector centrality varies significantly, ranging from 0.06 to 0.27, highlighting how eigenvector centrality captures not just the number of connections but also the influence of those connections. For instance, the vertex in the upper left, with an eigenvector centrality of 0.06, is connected to a less influential vertex with a centrality score of 0.2. In contrast, the vertex in the lower right has a higher eigenvector centrality of 0.27, as it is connected to a highly influential vertex with a centrality score of 0.95. This demonstrates that eigenvector centrality reflects the importance of a vertex’s connections, accounting for both the number and the significance of the connected vertices.

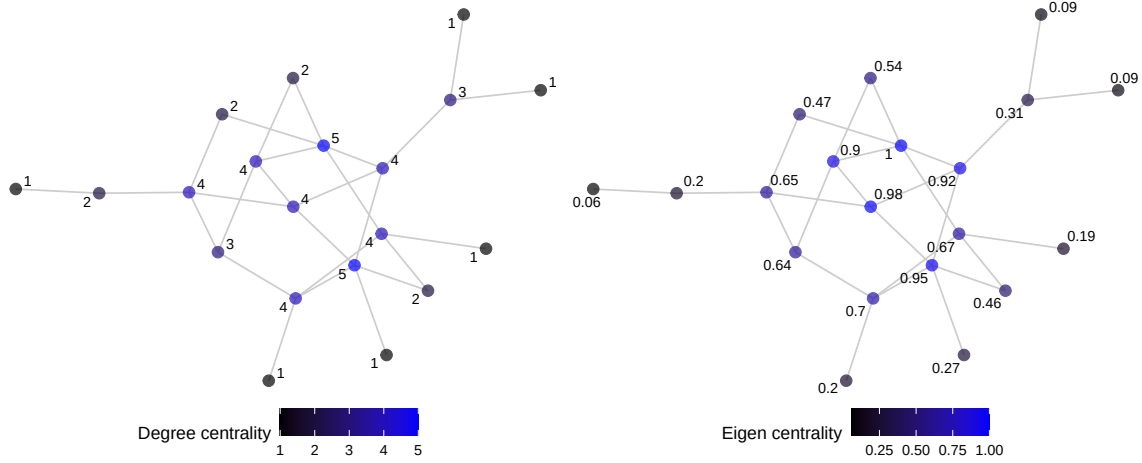


Figure 2: Example of a network with each vertex labeled by eigenvector centrality and degree centrality

To derive the eigenvector centrality of each legislative output, I construct a citation network for each policy field within each time period. This network captures the relationships between policies based on their citations, allowing for an assessment of the relative importance of each element within its field for a given year.

An important consideration when assessing impact is that its full significance can often only be evaluated ex-post. This contrasts with ex-ante approaches that measure legislative impact through immediate structural changes, such as amendments and republication authorizations (Garwe 2025). While these methods capture how a law reshapes the legal landscape upon enactment, they do not account for its evolving role in governance over time. Policies might not immediately show their influence, as their relevance is often revealed through later citations by subsequent policies.

Thus, rather than focusing solely on the initial legal modifications, this study examines how laws accumulate influence over time by tracking their position within the broader legal framework. For this, I will use several cut-off points ranging from 1 to 15 years after the policy's enactment to assess a policies impact. This approach avoids making premature decisions about when a legislative output's full impact becomes evident, providing flexibility and ensuring a more comprehensive analysis of its influence over both short- and long-term periods.

3.4 Measuring punctuated activity shifts

Turning to activity shifts, the most common approach in the literature to define them is to assess change rates in total number of outputs per period and field.

$$CR_{i,l,t} = \frac{P_{i,l,t} - P_{i,l,t-1}}{P_{i,l,t-1}} \times 100 \quad (3)$$

The change rate CR of a policy field i at a time point t is calculated as the difference between the policy output P at time t and the policy output at the previous time point $t - 1$, divided by the output at time $t - 1$. This result is then multiplied by 100 to express the change from the previous period as a percentage. The formula captures how much the policy output has increased or decreased compared to the previous period in percentage terms. I apply this change rate calculation across all policy fields.

Additionally, I account for the legal instruments l employed in policy making. In the context of EU policy-making, instrument choice is often a strategic and deliberate decision ([Hurka and Steinebach 2021](#)). Different instruments, such as regulations, directives, and decisions, embody distinct decision-making logics and legal effects, even within the same policy field. Regulations are directly applicable, directives require transposition into national law, and decisions target specific cases or actors, each shaping policy implementation in unique ways (see also [Haag, Hurka, and Kaplaner 2024](#)).

Thus, it is critical to differentiate not only between policy fields but also between the types of instruments used. By distinguishing between instruments, I capture variations in policy approaches that reflect the underlying strategic choices made by policy makers. This differentiation allows for a more nuanced analysis of how activity shifts manifest, as shifts in policy output may be driven not only by changes in the volume of activity but also by shifts in the types of instruments chosen, reflecting varying levels of EU integration, member state discretion, and sectoral priorities.

To define punctuated activity shifts — in the quantitative literature this is often the sole measure to identify punctuations — based on these change rates, the literature offers several approaches. One method is using a fixed cut-off point ([John and Bevan 2012](#)), another is relying on a specific quantiles of changes ([Mortensen 2009](#)), and a third involves superimposing a normal distribution over observed change values to identify outliers ([Robinson et al. 2007](#)).

Each method has its strengths and weaknesses. Defining extreme shifts by fixed values is relatively straightforward, but it may overlook nuances in fields with differing levels of policy output. For instance, in areas with limited policy activity, even small changes may represent significant shifts. Conversely, fields with consistent output may be less prone to extreme deviations, making the fixed-value approach less effective. While applying a normal distribution addresses these limitations and aligns with the theoretical underpinnings of PET, it is a more complex procedure.

In contrast, I opt for capturing punctuated periods through quantiles. This approach aligns well with the growing literature on quantile regression for explaining policy punctuations and has a mathematical connection to widely used measures of punctuation such as L-kurtosis and the Gini coefficient (Kaplaner and Steinebach 2022), which can be expressed through quantile spacing or quantile differences respectively.

To construct a binary variable indicating whether a legislative output occurred during a punctuated period, I use different quantile cut-off points of the change rate $CR_{i,t,l}$ within each policy field. Initially, I define punctuated periods as those where the change rate exceeds the 95th percentile of $CR_{i,l}$ within each policy field and legal instrument, identifying the most extreme policy shifts. The binary variable $Punctuated_{i,t,l}$ is defined as:

$$Punctuated_{i,t,l} = \begin{cases} 1 & \text{if } CR_{i,t,l} > Q_{.95}(CR_{i,l}) \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

where $Q_{.95}(CR_{i,l})$ represent the 95th cut-off points for the change rate in policy field i for a given legal instrument l .

To refine the analysis and account for temporal variations, I extend this criterion by applying the 95th to change rates within each field and decade. In this case, the binary variable $Punctuated_{i,t,l,\text{decade}}$ is calculated as:

$$Punctuated_{i,t,l,\text{decade}} = \begin{cases} 1 & \text{if } CR_{i,t,l} > Q_{.95}(CR_{i,l,\text{decade}}) \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

where $Q_{.95}(CR_{i,l,\text{decade}})$ and represent the 95th percentiles for the change rate within each policy field i , legal instrument l and decade.

This extension allows for a more dynamic analysis of policy punctuations, reflecting shifts in policy trends over time within each field. Figure 3 shows the resulting change rates across all policy fields in the EU with blue marking punctuated periods. In this case with the 95th per decade cutoff criterion. The calculations are based on the data presented in Section 3.5.

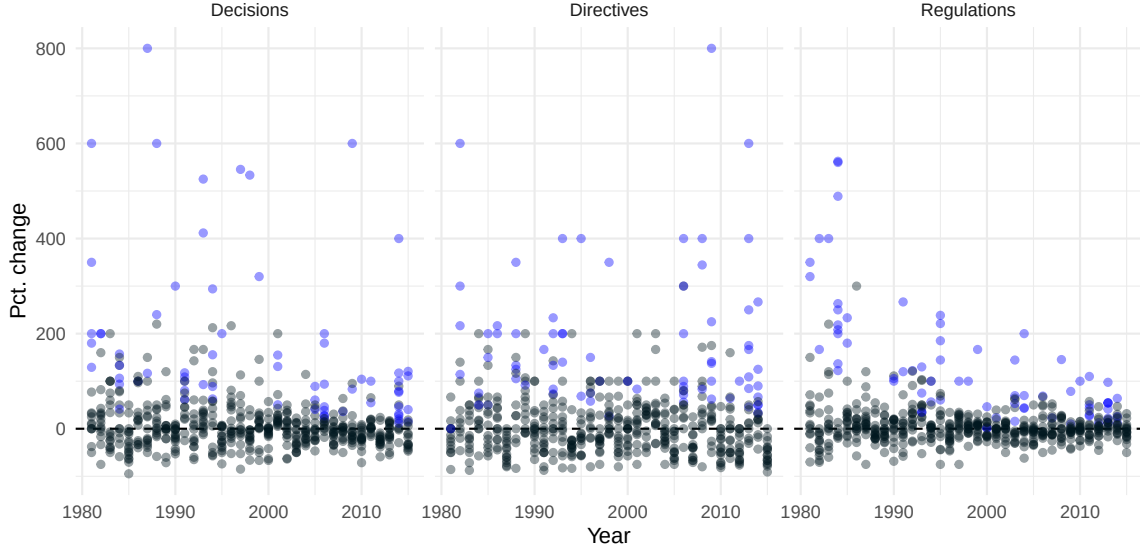


Figure 3: Change rates of all policy fields across time split by legal instrument. Punctuated periods defined as the 95th quantile per decade marked in blue.

3.5 Data

For the measurement and analysis I combine several data sources. For the network analysis I use data from Fjelstul (2019), the raw texts of the legislation used for the article embeddings are taken from Borrett and Laurer (2020) while the classification in policy fields based on EuroVoc is taken from Kaplaner, Knill, and Steinebach (2025). Overall, there are some prerequisites for the data to be included in the analysis. First, since the measures are always taken on the policy field level it is necessary for each law to contain EuroVoc indicators for the classification. Hence I focus on legislation stemming from 1980 onward since before that EuroVoc classification is often missing (see also Kaplaner, Knill, and Steinebach 2025). From the relevant policy fields I remove *Science* since it has comparatively low output, making the change rates rather erratic. Additionally, because Fjelstul (2019)’s dataset only includes data up to 2015, I limit the analysis to legislation between 1980 and 2015. This period ensures both the availability of EuroVoc classifications and the relevant network

data needed for this study. In total this results in 55,344³ legal documents relevant for the analysis with 161,111 unique articles. It is important to note, that one legislation can belong to multiple policy fields (see also [Kaplaner, Knill, and Steinebach 2025](#)). Hence, the number of observations is higher in the models reported below. An overview of all documents by policy field can be seen in the Supplementary Information (SI). Each document is split into individual articles to assess the innovation dimension. Notably, the last article is always excluded, as it typically specifies the applicability date of the provision, which offers limited insight into the policy’s innovativeness. Furthermore, I exclude articles from before 1985 because there are not enough years of prior legislation to ensure sufficient comparisons in the calculation of the innovation score. This leaves 344,768 articles for the analysis.

3.6 Empirical approach

The connection between legislative significance and policy punctuations is explored through a two-level approach, starting at the micro level and extending to the macro level.

First, the theory is tested at the micro level, focusing on individual provisions, or articles. This level is crucial because it allows for a detailed examination of whether punctuated periods produce outputs with distinct characteristics, such as higher innovation or greater long-term impact. By analyzing individual provisions, it is easier to control for intervening factors such as policy domain, legal instrument, or the specific context of a legislative output, ensuring that observed patterns are directly tied to punctuated periods rather than external influences. Starting at this level provides a clearer and more precise understanding of how policy making changes during bursts of activity.

Building on these micro level insights, the analysis then shifts to the macro level, aggregating innovation and impact scores over time. This perspective aligns with PET’s focus on systemic change, assessing whether punctuated periods in sum result in significant shifts in the broader legislative landscape. However, as cumulative patterns inherently depend on individual outputs, starting with the micro level ensures that aggregate trends are grounded in robust, context-sensitive findings.

³For the impact measures, there is some variability in the number of documents included in the analysis. A law must “survive” for a certain period to have its impact assessed, meaning the number of relevant documents decreases over time. For instance, the number of legislations that survived at least one year is 51,550. This issue is further elaborated in [Section 4](#).

3.7 Model specification

I will employ a linear fixed effects regression model with innovation and impact as the dependent variables (DVs), and policy punctuations as the independent variable (IV), along with a set of fixed effects. The main models are estimated using OLS. Additionally, I will use cluster-robust standard errors at the document level to account for potential within-group correlation. This is necessary since one legislative output can fall within multiple policy fields when it addresses issues from multiple policy domains (see also [Kaplaner, Knill, and Steinebach 2025](#)).

The fixed effects regression model can be specified as follows:

$$Y_i = \beta_0 + \beta_1 Punctuated_{i,t,l} + \delta_l + \alpha_i + \gamma_t + \zeta_d + \epsilon_{i,t,l,d} \quad (6)$$

Where:

- Y_i represents the **innovation** or **impact** score for a policy from field i .
- $Punctuated_{i,t,l}$ is a binary variable indicating whether policy field i at time t experienced a **policy punctuation** within legal instrument l (i.e., if the change rate exceeded the 95th percentile).
- β_1 represents the association between a **policy punctuation** and the **innovation** or **impact** score, reflecting how the score tends to vary when a punctuation occurs.
- δ_l represents the **fixed effects** for legal instrument l , controlling for instrument specific variation.
- α_i represents the **fixed effects** for policy field i , controlling for time-invariant characteristics specific to each policy field.
- γ_t represents the **time fixed effects**, accounting for factors that vary over time but are constant across policy fields (e.g., general political or economic conditions).
- ζ_d represents the **amending fixed effect**, accounting for differences in legislation that simply amend and legislation that constitutes new rules.
- $\epsilon_{i,t,l,d}$ is the error term, with **cluster-robust standard errors** at the document level to correct for heteroskedasticity and within-document correlations.

Table 1: Regression results for innovation

Dependent Variables: Model:	Innovation (field) (1)	Innovation (total) (2)	Innovation (field) (3)	Innovation (total) (4)
<i>Variables</i>				
Punctuation 95th (decade, field)	4.43×10^{-5} (0.0009)	0.0012 (0.0009)		
Punctuation 95th (field)			-0.0024** (0.0012)	-0.0003 (0.0011)
<i>Fixed-effects</i>				
Year	Yes	Yes	Yes	Yes
Legislative Instrument	Yes	Yes	Yes	Yes
Policy field	Yes	Yes	Yes	Yes
Amending	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	344,768	344,768	344,768	344,768
R ²	0.16304	0.11400	0.16307	0.11398
Within R ²	3.42×10^{-8}	2.66×10^{-5}	3.77×10^{-5}	5.69×10^{-7}

Clustered (Document) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

4 Results

Table 1 presents the results of the regression models assessing the innovativeness of individual provisions. Models (1) and (2) use within-field innovation scores, while Models (3) and (4) expand the comparison to include all legislation. In terms of dependent variables, Models (1) and (2) apply the decade- and field-specific punctuation measures, whereas Models (3) and (4) focus solely on the field-specific punctuation measure.

The results indicate that provisions from punctuated periods do not show a clear or consistent association with higher levels of innovation in individual provisions. In fact, Model (3) suggests a potentially negative association, where articles introduced during these periods may exhibit slightly lower levels of innovation. However, this finding should be interpreted with caution, as the negative association is not robust across all models and the magnitude of the effect appears modest.

These results offer tentative support for the theory that, during times of heightened attention and significant policy shifts, policy makers might be inclined to default to more incremental or familiar approaches rather than adopting truly innovative measures. As suggested before, this could be explained by the pressure to act quickly or to address immediate crises, leading to a reliance on known policy solutions rather than the pursuit of new and experimental ideas. Consequently, individual provisions from punctuated periods do not appear to significantly

Table 2: Regression results for impact

Dependent Variable:		Impact							
Impact after:		1 year	5 years	10 years	15 years	1 year	5 years	10 years	15 years
Model:		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Variables</i>									
Punctuation 95th (decade, field)		0.0098*** (0.0006)	0.0064*** (0.0009)	0.0038*** (0.0009)	0.0016* (0.0008)				
Punctuation 95th (field)						0.0196*** (0.0015)	0.0137*** (0.0015)	0.0017* (0.0009)	0.0044*** (0.0011)
<i>Fixed-effects</i>									
Policy field		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Legislative Instrument		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Amending		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>									
Observations		108,317	66,193	35,942	17,379	108,317	66,193	35,942	17,379
R ²		0.08614	0.08120	0.07701	0.08123	0.08794	0.08320	0.07631	0.08216
Within R ²		0.00731	0.00258	0.00085	0.00028	0.00927	0.00476	8.79×10^{-5}	0.00130

Clustered (Document) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

differ in their innovativeness compared to those from non-punctuated periods. This suggests that the presence of a punctuation does not necessarily lead to increased innovation at the granular level of specific legal provisions. Instead, punctuated periods might reflect a more cautious, incremental approach to policy formulation.

Table 2 shows the result from the same specification from above for the impact measure. As before, I rely on the different punctuation definitions. Additionally, the models use the impact measure at different time points varying from 1 to 15 years. It is important to note here, that the later the impact is assessed, the lower the N becomes. This is due to the fact that a legislation has to survive for, for example, 15 years to be assessed in terms of its impact at this time point. This reduces the viable observation in two ways, first, it reduces the sample to legislation to at least 15 years old, second, it removes legislation abolished within this period.

The results reveal a slightly different pattern than previously observed. Across all models, there is a modest positive association between punctuated periods and impactful legislation. Interpreting the effect sizes, particularly in model (6), which represents the impact after 5 years as shown in Table 2, suggests that outputs from punctuated periods are associated with an increase in impact equivalent to approximately 114% of the average impact score. However, as shown in Figure 4, the differences between legislation passed during punctuated periods and non-punctuated periods appear to diminish over time.

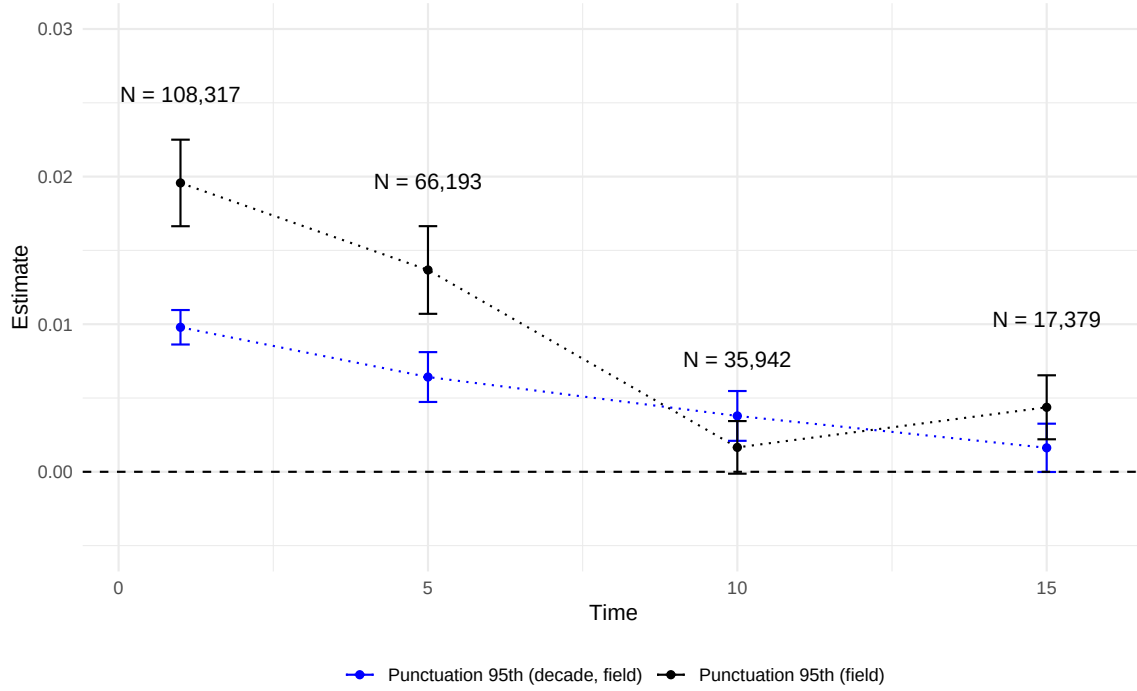


Figure 4: Punctuation coefficients for different time points of assessing Impact with 95% confidence intervals. Note: The estimation of these coefficient are based on different samples and consequently derived from different models.

There are two potential explanations for this trend. First, as previously noted, punctuated periods are characterized by stark attention shifts, often resulting in a surge of legislative activity. This heightened productivity during such periods could lead to an initial increase in the impact of legislation, as more laws are enacted, many of which likely address similar or interconnected topics. This interconnectedness may initially boost the perceived impact of legislation passed during these times compared to quieter periods.

Second, the observed differences in impact may diminish over time due to a form of selection bias. As time progresses, impactful legislation — regardless of whether it was passed during a punctuated or non-punctuated period — is more likely to remain relevant and endure. As a result, when examining legislation that has been in place for over 15 years, we may simply be comparing impactful laws to other impactful laws, irrespective of the period in which they were enacted. This raises an important question: Does legislation from punctuated periods have a higher likelihood of long-term survival?

Table 3 addresses this question by modeling survival time using a similar specification as before, but with survival time as the dependent variable and a poisson instead of a linear

Table 3: Poisson regression results for survival. Note: The N is slightly lower as in Table 2, provision that are more or less immediatly abolished are excluded.

Dependent Variable: Model:	Survival time	
	(1)	(2)
<i>Variables</i>		
Punctuation 95th (decade, field)	0.0100** (0.0050)	
Punctuation 95th (field)		0.0195** (0.0084)
<i>Fixed-effects</i>		
Legislative Instrument	Yes	Yes
Year	Yes	Yes
Censored	Yes	Yes
Amending	Yes	Yes
Policy field	Yes	Yes
<i>Fit statistics</i>		
Observations	108,260	108,260
Squared Correlation	0.70791	0.70787
Pseudo R ²	0.34079	0.34080
BIC	613,477.1	613,469.7
<i>Clustered (Document) standard-errors in parentheses</i>		
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

regression model. Survival time here reflects how long a policy remained in effect before being repealed or replaced, with the endpoint set as 2015. To account for censoring (i.e., policies still in effect by 2015), I include an additional fixed effect to adjust for policies that have not yet reached an endpoint.

The results reveal a significant positive impact on survival, suggesting that legislative outputs from punctuated periods tend to last slightly longer than those from non-punctuated periods. This finding indicates that legislation enacted during these high-attention periods may indeed be more enduring, supporting the association that punctuated periods not only produce slightly more impactful legislation in the short term but also legislation with slightly higher long-term staying power. Nevertheless, the differences are rather modest.

5 Categorizing policy punctuations

So far, I have focused on how well legislative activity aligns with innovation and impact. When it comes to classifying punctuations, we can treat these three dimensions — activity, innovation, and impact — as distinct but interconnected components. First, activity shifts can be viewed as the baseline indicator of policy punctuations. Without shifts in activity, punctuations are unlikely to occur, as the literature on PET suggests that changes in attention and political responsiveness manifest primarily through surges in policy production or legislative output.

The findings of this study challenge the assumption that increased activity alone is sufficient to result in innovation or generate significant legislative impact. Although activity shifts are a necessary precondition for punctuations, they do not automatically translate into groundbreaking or impactful policy change. This is in line with critiques within the PET literature, which suggest that not all attention shifts lead to substantive policy transformation. As John and Bevan (2012) argue, high legislative activity can result from a variety of processes, including procedural expansions, low-salience regulatory adjustments, or high-salience responses to external shocks. This aligns with findings in my analysis, where bursts of activity — while necessary to signal a punctuation — often correlate with low levels of innovation and are only weakly associated with increased impact.

Thus, while activity shifts might be necessary to classify policy punctuations, they are not sufficient in themselves to capture the full extent of policy change. Punctuations that reflect significant shifts in the policy landscape must also exhibit high levels of either innovation, impact, or both. When categorizing punctuations, it becomes crucial to evaluate whether increased legislative activity is accompanied by transformative innovation (i.e., introducing new ideas or regulatory frameworks) and whether these changes have lasting impact (i.e., reshaping the policy system and influencing future legislative outputs).

In this light, policy punctuations can be better understood by analyzing the intersection of activity with innovation and impact, leading to a more nuanced classification that distinguishes between different types of punctuations. For instance, while all punctuations involve heightened activity, the degree to which this activity is coupled with innovation and impact determines whether the punctuation represents a superficial burst of output or a deeper structural change. This approach moves beyond the traditional focus on activity shifts alone and incorporates the substantive quality of policy change into the assessment,

providing a more comprehensive understanding of the dynamics at play during periods of intense legislative attention.

By integrating these three dimensions we can better understand the types of policy changes that occur during punctuated periods. Not all punctuations are alike, and not all bursts of legislative activity produce transformative outcomes. Some may result in groundbreaking innovations, while others may simply consolidate existing frameworks. To illustrate this variability, I introduce a typology of policy punctuations based on their levels of innovation and impact.

5.1 Classifying punctuations

This typology distinguishes four types of policy punctuations, each representing a different interaction between legislative activity, innovation, and impact:

- **Transformational Punctuations** (High Activity + High Innovation + High Impact): These are marked by significant legislative activity, coupled with innovation and lasting systemic impact. They lead to structural shifts that redefine the trajectory of a policy domain. They result in profound, long-term changes in governance and policy systems.
- **Consolidating Punctuations** (High Activity + Low Innovation + High Impact): This type of punctuation represents periods of high legislative activity focused on consolidating existing frameworks, reinforcing what is already in place, and addressing emerging challenges. While not particularly innovative, the policies implemented during these periods have lasting, high impact on maintaining or improving policy coherence and resilience.
- **Experimental Punctuations** (High Activity + High Innovation + Low Impact): These are characterized by bursts of creativity and innovation but lack significant long-term impact. They represent experimental policy shifts or attempts to introduce new ideas, but the changes are often short-lived or do not fully integrate into broader frameworks, leaving limited long-term systemic change.
- **Procedural Punctuations** (High Activity + Low Innovation + Low Impact): These periods reflect high levels of legislative activity, but the policies produced are routine or incremental, lacking in significant innovation or long-term impact. The activity

might be influenced by administrative requirements, procedural obligations, or symbolic gestures, which may not necessarily reflect substantive policy shifts.

This classification scheme highlights the variability in the outcomes of policy punctuations. Not all legislative surges result in profound or meaningful reforms. Some are innovative but have little impact, others consolidate existing systems without introducing new ideas, and some produce routine outputs with no significant consequences.

5.2 Application to the EU Case

Next, I apply this categorization to the case of the EU. For this, I no longer look at individual provisions but instead focus on shifts in overall *Activity* (percentage changes per field and year), *Innovation* (summed per field/year), and *Impact* (summed per field/year)⁴. All values are represented as per-field z-scores to facilitate comparability.

I divide the space according to the four classifications. To identify “high” values, I apply a threshold of 1.4 standard deviations above the mean. While the precise quantile this corresponds to varies somewhat across variables due to distributional differences, it generally falls between the 90th and 95th percentiles.

⁴For the *Impact* measure, I assess legislative influence over a five-year period following enactment. Similarly, *Innovation* is evaluated using a five-year reference window. To reduce complexity, I focus on *Innovation* relative to all legislative outputs.

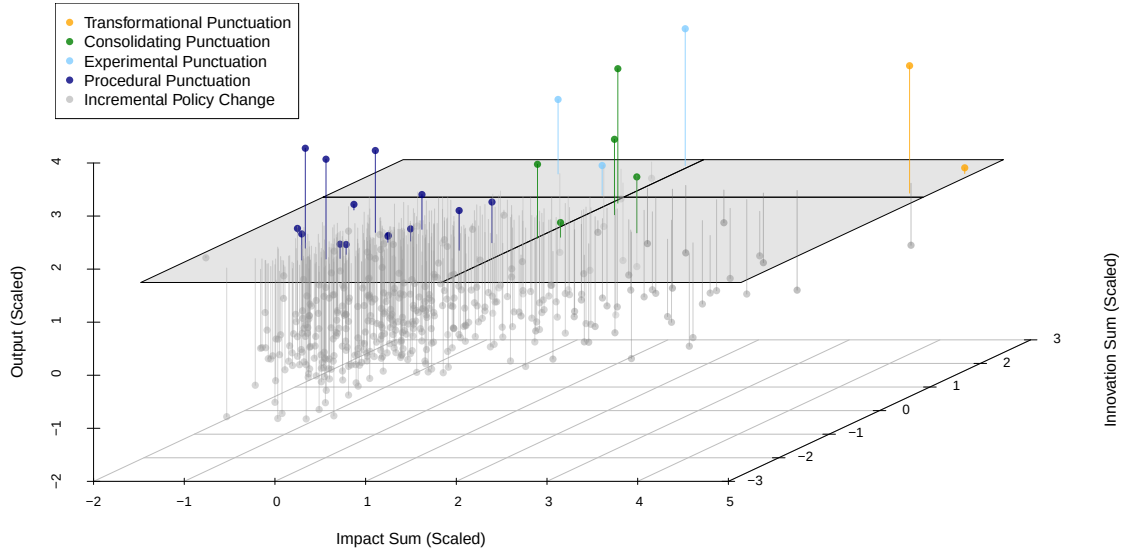


Figure 5: Policy change categorization for the EU.

Figure 5 illustrates the classification results, with *Activity* on the *y*-axis, *Impact* on the *x*-axis, and *Innovation* on the *z*-axis. The grey plane marks the threshold for what would be considered a punctuation based on *Activity*. Points above it represent activity shifts that can be considered as punctuations. The lines dividing the plane indicate the cut-off points for high *Innovation* and *Impact*. This creates four quadrants: Transformational Punctuations in the upper right, Consolidating Punctuations in the lower right, Experimental Punctuations in the upper left, and Procedural Punctuations in the lower left.

If shifts in legislative activity were a reliable indicator of innovation and impact in legislative outputs, we would expect most values to cluster in the upper-right corner. However, consistent with previous results at the individual provision level, it is unsurprising that many punctuations in Figure 5 deviate from this pattern⁵. Instead, when categorized by *Innovation* and *Impact*, a significant number fall into the Volume-Driven Punctuation category. Additionally, in line with earlier findings, Consolidating Punctuations are slightly more common than Experimental Punctuations.

Notably, there are only two punctuations that represent a Transformational Punctuation: Economic policy in 1994 and environmental policy in 1993⁶. Both cases are marked by high

⁵See also the patterns for aggregated means divided by legal instrument in the SI.

⁶For examples for the other types of punctuations see the SI.

activity, high innovation, and high impact, exemplifying profound shifts in their respective policy domains.

A prime example of such a *Transformational Punctuation* is economic policy in 1994, featuring a surge in legislative output alongside profound restructuring of the EU’s economic governance. That year saw the second stage of EMU launch on January 1, creating the European Monetary Institute — a precursor to the ECB — and included major acts such as Directive 94/19/EC (harmonizing deposit-guarantee schemes) and Council Regulation (EC) No 1164/94 (establishing the Cohesion Fund). These initiatives enhanced financial stability and cohesion, laying crucial groundwork for the euro (Verdun 2000). Additional legislation — like Council Regulation (EC) No 3281/94 (opening tariff quotas), Directive 94/22/EC (hydrocarbon licensing), and Directive 94/45/EC (European Works Councils) — underscored the scope and systemic impact of the 1994 reforms.

Similarly, environmental policy in 1993 marked another *Transformational Punctuation*, shaped by the Maastricht Treaty’s emphasis on mainstreaming environmental protection (Lenschow 2002; Knill and Liefferink 2021). While the Fifth Environmental Action Programme was introduced via a Council Resolution rather than a binding act, it nonetheless provided a strategic framework that inspired a series of important legislative measures. Among these was Council Regulation (EEC) No 792/93, which established a temporary cohesion financial instrument to support environmental projects in Member States with a per capita GNP below 90% of the EU average. This regulation prioritized funding for environmental protection initiatives, complementing broader efforts to reduce disparities between regions. Several accompanying Council decisions implemented this instrument throughout 1993, channeling resources toward projects that addressed environmental challenges in the EU’s less developed regions.

In conclusion, while economic policy in 1994 reshaped the EU’s fiscal and monetary architecture, environmental policy in 1993 laid down a comprehensive framework for sustainable governance. Both cases exemplify high-activity, high-innovation, and high-impact punctuations, illuminating their enduring significance in European policy-making.

6 Conclusion

This paper set out to investigate whether policy punctuations — periods of elevated legislative activity — translate into more innovative or impactful legislation within the EU.

Overall, the findings offer a more nuanced perspective on PET, highlighting five main observations:

First, legislative bursts do not reliably foster innovation. Contrary to assumptions that punctuated periods might bring more novel ideas, the evidence suggests no consistent increase in innovation levels. In some instances, the data even point to slightly less innovation during these intensified periods, suggesting a propensity for incremental rather than ground-breaking solutions.

Second, punctuations show only a modest increase in long-term impact. Laws passed during these bursts do display a tendency impact subsequent legal frameworks slightly more. However, this effect is far from definitive and should not be interpreted as a rule. The overall influence of punctuated legislation appears scattered across different issues and time spans.

Third, macro-level patterns can be misleading if one focuses solely on the number of laws. Raw output alone often conceals the substantive quality of legislation, making it difficult to discern whether a sudden flurry of acts consists largely of minor adjustments or genuinely transformative measures.

Fourth, qualifying legislative outputs by examining both innovation and impact is crucial. By looking at the underlying content and the degree to which laws influence future policy making, it becomes clear that not all bursts of activity are equally consequential. This underscores the importance of examining what is being legislated, rather than simply how much.

Finally, genuinely transformative punctuations prove rarer than raw data would suggest. Once one distinguishes between nominal upticks in legislation and actual transformational change, many supposedly momentous episodes turn out to be more of the same rather than deep policy overhauls.

One possible explanation for these patterns is that many policy punctuations remain within policy subsystems, limiting their potential for broader systemic change. While PET suggests that major shifts can occur when issues break free from subsystem constraints and reach the macro-political agenda, this may be relatively rare. Instead, many punctuations likely reflect technocratic or administrative adjustments rather than fundamental policy reconfigurations. Without broader political engagement, heightened legislative activity may reinforce existing frameworks rather than disrupt them.

This could help explain why only a few punctuations—such as the 1994 economic policy shift surrounding the Euro or the 1993 environmental policy reforms—produced genuinely transformative change. These cases saw issues gaining political salience, attracting new actors, and disrupting pre-existing policy monopolies, resulting in lasting institutional shifts. By contrast, many other punctuations, despite high legislative activity, remained within subsystem boundaries, leading to incremental adjustments rather than sweeping reforms.

Taken together, these insights indicate that while punctuated periods may sometimes yield impactful or innovative laws, they do not invariably bring forth novel or high-impact change. Future scholarship could further clarify the specific institutional or external conditions that enable genuinely transformative policy-making during surges of legislative activity. In so doing, it would refine our understanding of when and how punctuations reshape the EU’s legislative landscape in a truly meaningful way.

Beyond these substantive conclusions, this research also demonstrates how quantitative methods can help “crack open” punctuations and evaluate their quality at scale. By embedding legislative text to measure innovation and constructing citation networks to assess impact, the study offers replicable, data-intensive techniques that can be applied to large legislative corpora. These methods enable a more precise distinction between punctuations that foster enduring reforms and those that merely expand legislative output, thereby opening new avenues for understanding the conditions under which policy bursts produce genuinely transformative change.

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Supplementary Information

A.1 Aggregated impact and innovation score against change rate per legal instrument.

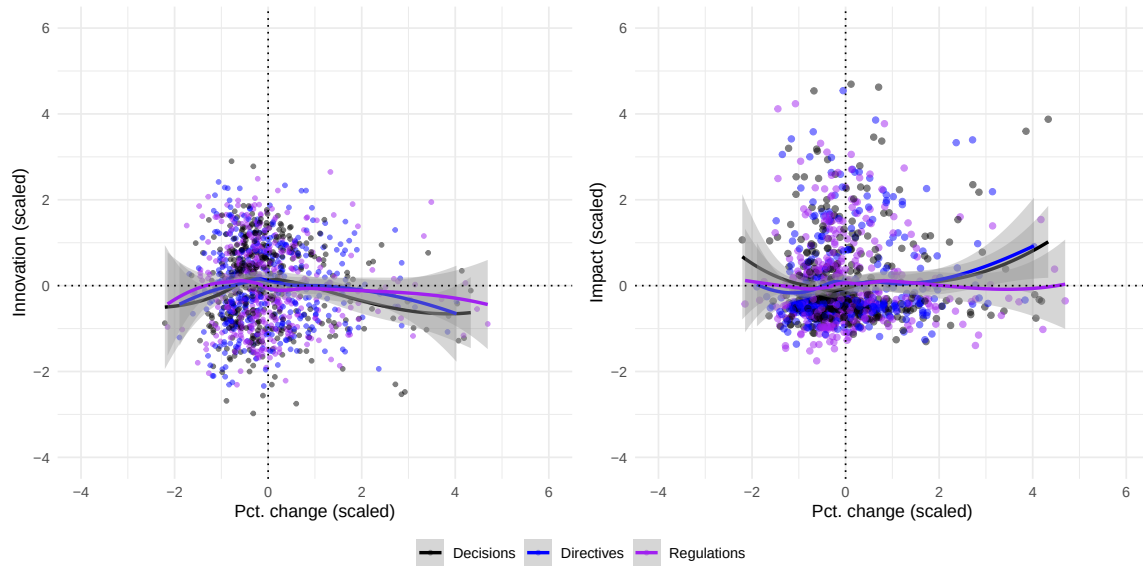


Figure A.1: Mean yearly impact (5 year) and innovation scores (scaled) for each policy field and legal instrument, plotted against the rate of change. The regression lines are fitted using LOESS to illustrate trends and relationships between the variables

A.2 Aggregated innovation score against change rate per policy field per legal instrument.

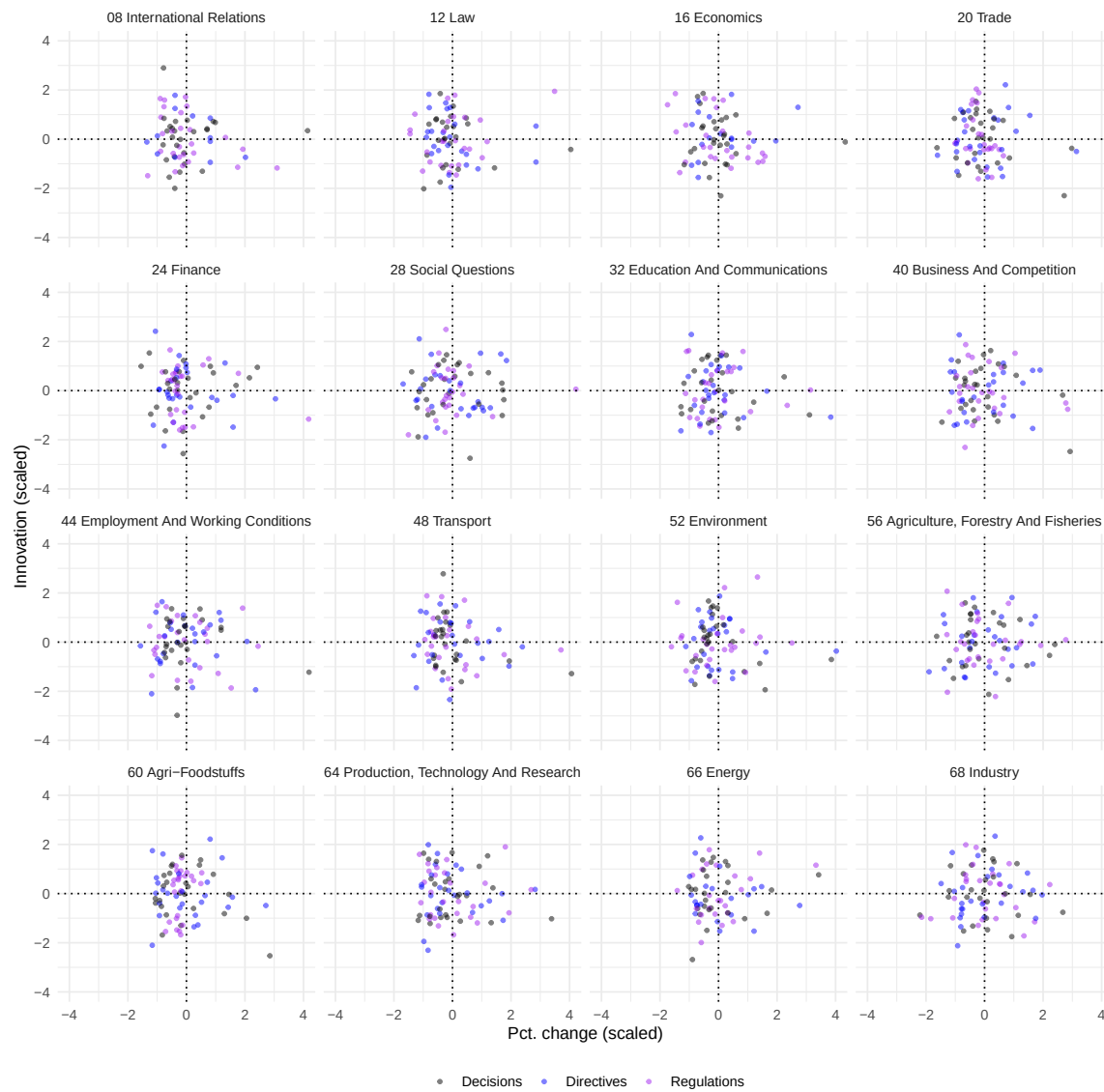


Figure A.2: Mean yearly innovation scores (scaled) for each policy field and legal instrument, plotted against the rate of change.

A.3 Aggregated impact score against change rate per policy field per legal instrument.

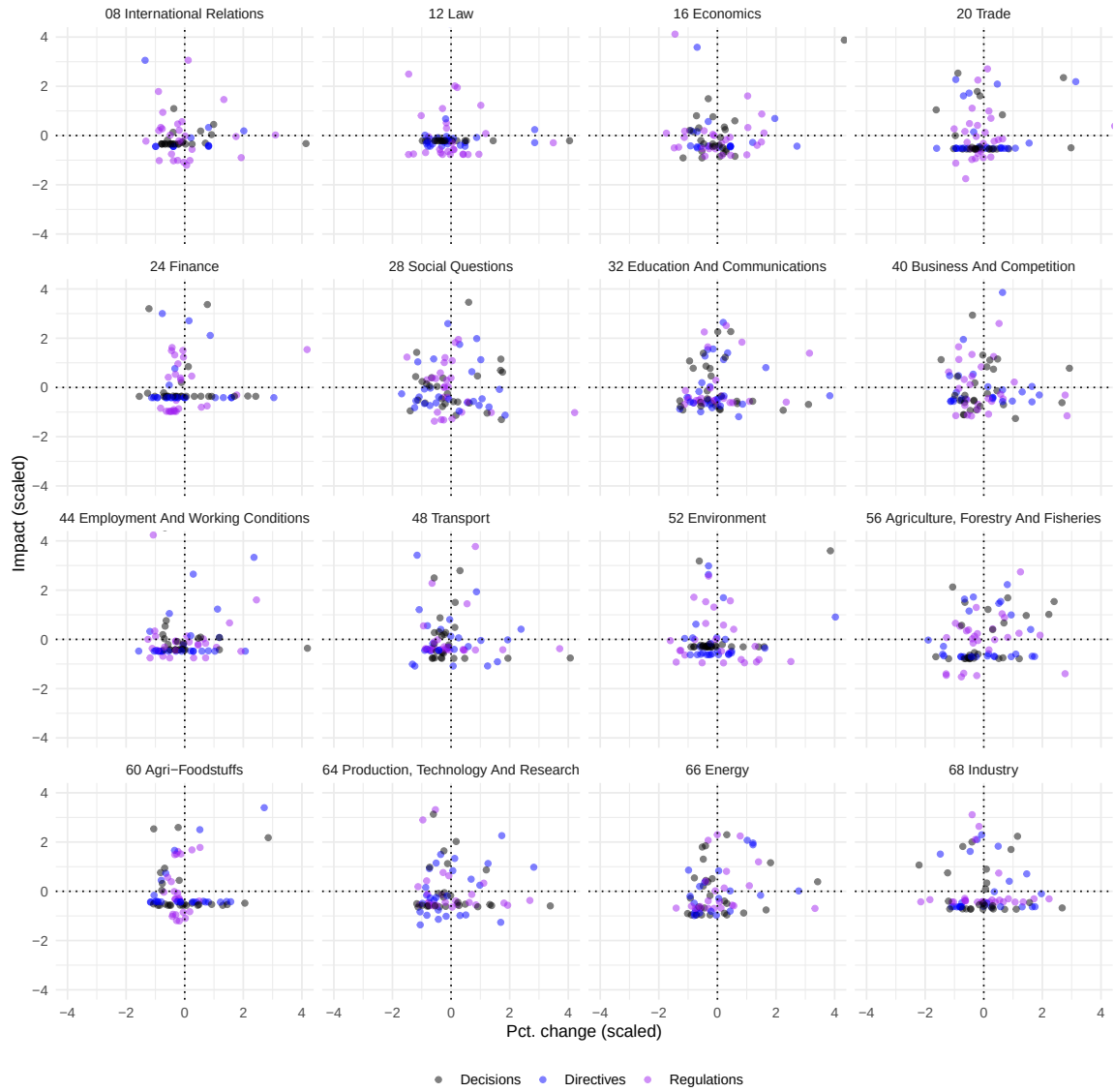


Figure A.3: Mean yearly impact (5 year) scores (scaled) for each policy field and legal instrument, plotted against the rate of change.

A.4 Distribution of innovation score and impact.

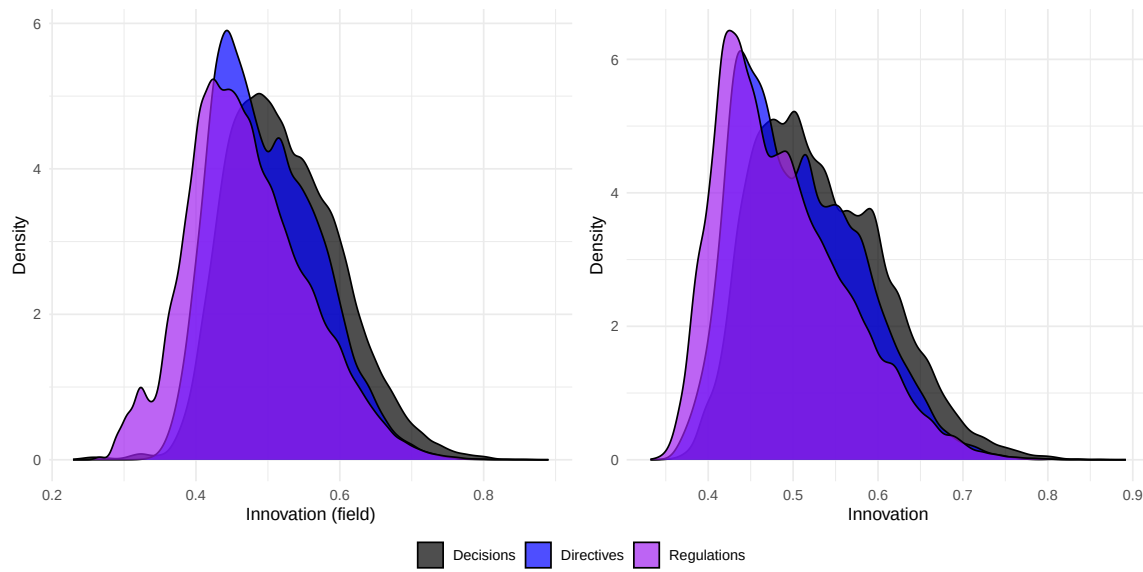


Figure A.4: Density distribution innovation score.

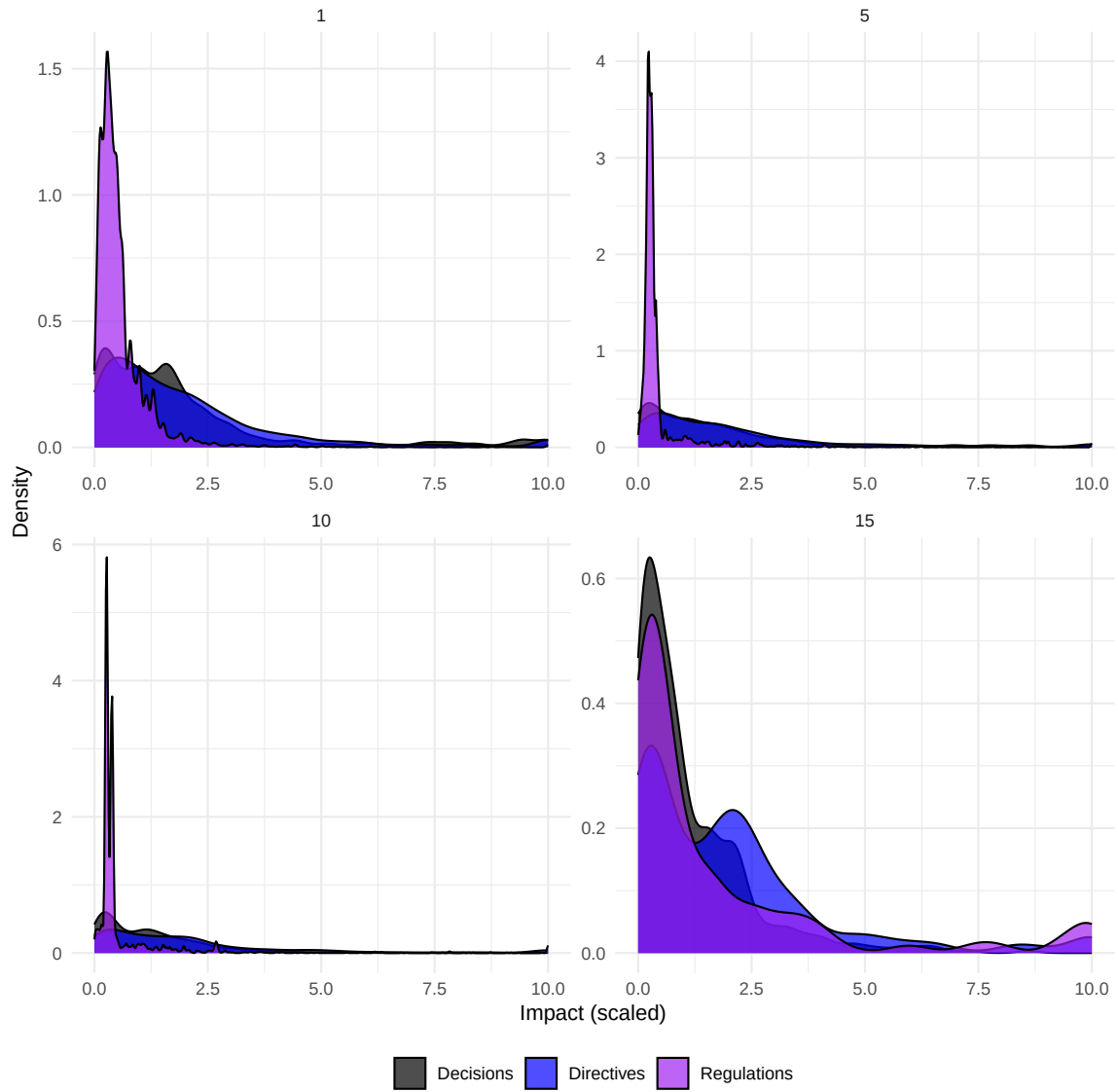


Figure A.5: Density distribution impact score for multiple reference time point. The score was scaled to ease comparison between reference points. Observations with a score higher than 10 were bundled at 10. Observations with a score of 0 were removed.

A.5 Innovation per policy field over time.

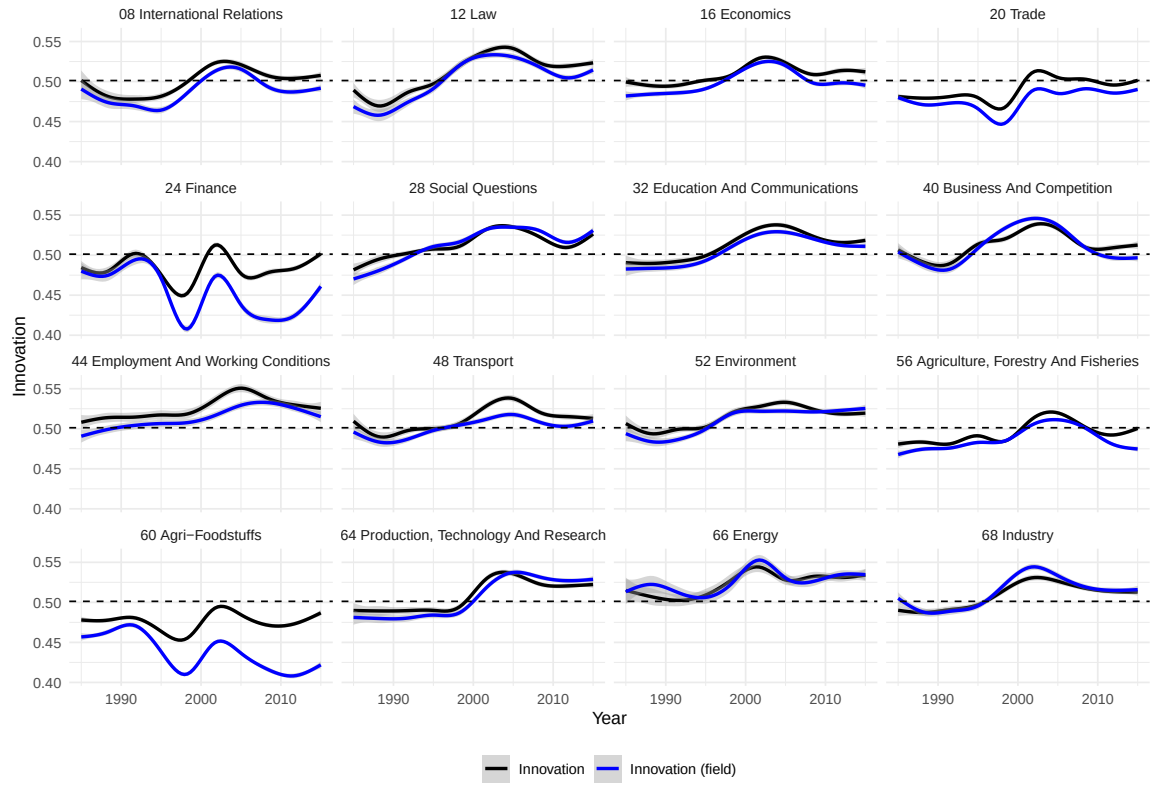


Figure A.6: Innovation scores for each policy field against years. Trend line fitted using GAM. Dotted line represents mean Innovation (total) score across all observations.

A.6 Overview included documents

Table A.1: Included documents per legal instrument and policy field

Field	Decisions	Directives	Regulations
08 International Relations	785	44	2982
12 Law	444	204	1019
16 Economics	2913	68	3561
20 Trade	3779	1233	28112
24 Finance	1014	280	8490
28 Social Questions	3019	899	2135
32 Education And Communications	1315	383	1736
36 Science	256	132	98
40 Business And Competition	2255	130	1569
44 Employment And Working Conditions	432	240	405
48 Transport	890	535	1153
52 Environment	1229	710	1024
56 Agriculture, Forestry And Fisheries	4901	891	13700
60 Agri-Foodstuffs	1855	459	20765
64 Production, Technology And Research	1301	628	1874
66 Energy	346	135	384
68 Industry	1914	545	2316

A.7 Example for other punctuations

A.7 .1 Experimental punctuation: Employment and working conditions in 2006

Employment and working conditions in 2006 marked an experimental phase of legislative innovation — a period of high policy ambition but limited long-term entrenchment. Key measures, including Directive 2006/54/EC on gender equality in employment and Council Decision 2006/702/EC on the renewed Lisbon Strategy, introduced *flexicurity* and sought to modernize labor market governance by balancing flexibility with worker protections. While innovative in scope, these reforms struggled to embed themselves as foundational pillars of EU labor policy. Weak legislative linkages meant that few subsequent regulations directly built upon them, and their influence remained largely conceptual rather than structural. The 2008 financial crisis further deprioritized these initiatives, shifting EU focus toward economic stabilization and austerity measures. Although some flexicurity principles re-emerged in later debates, particularly in the European Pillar of Social Rights in 2017, the 2006 reforms ultimately did not catalyze a lasting transformation of EU labor governance.

A.7 .2 Consolidating punctuations: Trade, Finance, and, International Relations in 1995

In 1995, the EU adopted Council Decision 95/1/EC to formalize the accession of Austria, Finland, and Sweden — which led to consolidating punctuations in the field of Trade, Finance, and International Relations. While enlargement significantly expanded the Union’s membership, it primarily reinforced rather than reconfigured established frameworks. For instance, Council Regulation (EC) No 1488/95, which deepened the EU’s Mediterranean Partnership, extended existing diplomatic and economic cooperation without altering the Union’s external engagement strategy. Likewise, the integration of new member states into the common trade regime maintained the EU’s liberalization trajectory rather than redefining it. In finance, Council Regulation (EC) No 2727/95, refining price stabilization under the Common Agricultural Policy, adapted existing mechanisms to a larger Union but did not overhaul financial governance. Thus, 1995 was high-impact in scope — broadening the EU’s institutional and economic base — yet low-innovation in that it consolidated rather than transformed the Union’s policy structures.

A.7 .3 Procedural punctuation: Energy in 1999

In 1999, the EU’s energy policy experienced a procedural punctuation, characterized by a surge in legislative activity that focused on regulatory fine-tuning rather than substantive transformation. This increase was largely driven by the implementation phase of Directive 96/92/EC, the EU’s first electricity market liberalization directive, which required secondary legislation to clarify technical and market rules rather than introducing new policy frameworks. Similarly, Directive 1999/32/EC, which regulated sulfur content in fuels, and Council Directive 1999/94/EC, which mandated consumer fuel economy labeling, primarily refined existing regulatory structures rather than initiating structural reforms or policy innovation.

While these measures contributed to harmonization and administrative consistency, their innovation levels were low, as they largely codified and operationalized prior commitments rather than introducing new regulatory paradigms. Their impact was also limited, as they lacked strong legislative linkages and enforcement mechanisms, reducing their ability to reshape the broader energy governance landscape. Moreover, with EU energy policy still fragmented across national jurisdictions, these procedural refinements failed to generate meaningful systemic integration.

Thus, 1999 exemplifies a procedural punctuation, where a temporary surge in legislative activity primarily served to clarify and standardize existing rules rather than drive transformative change. This case underscores that policy punctuations do not necessarily produce substantive policy shifts; instead, they may reflect bursts of administrative refinement that leave the overarching governance framework largely intact.